

HOJUNE KIM

hojune@stanford.edu | hojjunekim.github.io

EDUCATION

Stanford University

M.S. in Aeronautics and Astronautics

- GPA: **4.0/4.0**(Overall), Full-ride scholarship recipient

CA, United States

09/2025 – 03/2027 (Expected)

Seoul National University (SNU)

B.S. in Aerospace Engineering, summa cum laude

- GPA: **4.0/4.0**(Major), 3.91/4.00(Overall)

Seoul, South Korea

03/2019 – 08/2025

* 1.5-year absence from military service

WORK EXPERIENCE

Foundation Robotics

Robotics Research Intern

- Developing a generalized teleop-free loco-manipulation VLA training pipeline using world models, extending the embodied Gaussian Splatting approach from the LEGS paper to broader scenes and embodiments

CA, United States

06/2026 – 09/2026

German Aerospace Center (DLR) - Humanoid TORO team

Guest Researcher, Advisor: Prof. Jinoh Lee

- Implemented an adaptive locomotion system using semantic SLAM to optimize 8 gait parameters on uneven and deformable terrains via motion planning based on Divergent Components of Motion
- Developed a scalable semantic segmentation model in Jetson Orin AGX by fusing a stream-based active learning classifier with the Segmentation Anything Model, training it via a human-robot interaction loop using a HoloLens
- Improved resilience to common failures (e.g., contact loss, localization glitches) by implementing a Kalman Filter for robust visual-proprioceptive odometry, winning 1st place in the IEEE-RAS Humanoids competition

Munich, Germany

09/2024 – 02/2025

RESEARCH EXPERIENCE

Stanford University - Multi-robot Systems Lab

Research Associate, Advisor: Prof. Mac Schwager

- Designed a Gaussian Splatting simulator for loco-manipulation, enabling automated large-scale data generation
- Developing a zero-shot visuomotor policy for humanoids, eliminating dependence on human demonstrations

CA, United States

09/2025 – Present

ETH Zurich

Robotic Systems Lab, Research Associate, Advisor: Prof. Cesar Cadena

- Developed and trained reinforcement learning policies using a knowledge distillation framework for crowded navigation on the quadruped robot ANYmal, achieving a 74.5% success rate on static and dynamic terrains
- Utilized PPO in Isaac Lab for training and implemented a Gated Recurrent Unit to handle dynamic obstacles

Zurich, Switzerland

02/2024 – 06/2024

Vision for Robotics Lab, Research Associate, Advisor: Prof. Margarita Chli

- Developed a symbolic Gaussian Belief Propagation (GBP) solver in C++ for continuous-time SLAM, achieving a 2.4x to 110.3x speedup compared to conventional methods
- Benchmarked the performance of distributed optimization GBP against the centralized NLLS solver (Ceres) for both continuous- and discrete-time SLAM using hand-achieved data

02/2024 – 08/2024

Seoul National University - Robust Perception and Mobile Robotics Lab

Research Associate, Advisor: Prof. Ayoung Kim

- Proposed a novel 2D odometry method with only mmWave radar, achieving an 85-90% reduction in RPE on indoor datasets by using a bidirectional weighted ICP algorithm
- Designed a handheld multi-sensor system and held camera-LiDAR-radar joint calibration via graph optimization

Seoul, South Korea

02/2023 – 01/2024

PUBLICATIONS

- H. Kim**, T. Chen, J. Sun, L. W. Osterberg, Q. Chen, K. Wang, M. Schwager. LEGS: Fine-Tuning Teleop-Free VLAs for Humanoid Loco-manipulation in an Embodied Gaussian Splatting World. *CoRL* 2026.
- H. Kim**, H. Jang, A. Kim. 2D Ego-Motion with Yaw Estimation using Only mmWave Radars via Two-Way weighted ICP. *IEEE ICRA 2024 Workshop on Radar in Robotics*, Yokohama, 2024.
- H. Kim***, J. Lee*, G. Mesesan, R. Giubilato, R. Schuller, K. Fründ, J. Lee. Knowledge-enabled Adaptive Locomotion Using Semantic SLAM with Active Learning. *In Progress*.

SKILLS

Platforms: Humanoids, ANYmal, GO2, Drone, Vehicle

Frameworks: ROS1/2, Isaac Sim/Lab, MuJoCo, PX4, Gazebo

Programming: C/C++, Python, MATLAB

Tools: Cmake, AWS, Docker, Git, CAD

AWARDS

1st Place , IEEE-RAS Humanoids Competition Adult-sized Biped Free Walk	11/2024
Minister of National Defense Award (out of 953 teams), Korea Defense Startup Challenge	12/2022
Gold Prize (out of 60 teams), International Student Car Competition Autonomous Driving Sector	10/2021
SNU Alumni Association Chairman's Award (Sole recipient among the College of Engineering)	08/2025
• Awarded for exceptional leadership and contributions to the university community	

EXTRA CURRICULAR

Dentronic	07/2024 – 12/2025
<i>Co-Founder</i>	
• Built a back-drivable robotic assistant arm designed to automate suction and retraction for solo dentist procedures	
• Generated \$125k in revenue from the first product batch in a single day, serving customers across the US and Korea	
Fire Butterfly , Autonomous Drone Association	02/2023 – 08/2023
<i>Lead Software Engineer</i>	
• Engineered auto-landing system on drone with sub-5cm landing accuracy using C2 Bézier curve motion planning	
• Validated the system in both Gazebo simulation and PX4 on-board flight with Jetson Xavier, leading to a technology transfer to a private company	
Army Aviation Operations Command , Republic of Korea Army	08/2021 – 02/2023
<i>CH-47D Helicopter Flight Attendant, Sergeant</i>	
• Conducted maintenance on CH-47D helicopters and participated in tactical missions, including over 60 flight hours dedicated to two major forest fire extinguishment operations (2022 Uljin, 2022 Yeongdeok)	
AI Tech Play , Non-Profit Organization for AI education	02/2021 – 08/2021
<i>Co-organizer & Hardware Leader</i>	
• Collaborated with MIT Lincoln Lab to host AI camp and RC-car competition for over 200 nationwide children, providing RC-car simulation tools and handmade platform to students	
• Led a team to develop a multi-modal, ROS2-controlled RC-car with Jetson Nano, creating 7 Python tutorials on autonomous driving	
SNU ZERO , Autonomous Driving Car Association	01/2021 – 08/2021
<i>Sensor Technician</i>	
• Designed the autonomous vehicle system from chassis, developing GNSS-aided inertial navigation system via Extended-Kalman Filter assisted by line detection for robust localization	
• Engineered a multi-modal perception system for dynamic and static obstacle avoidance, using camera-LiDAR fusion	

PATENTS

1. Parking Location Tracking System, *KR102291377B1*, 2021
Hojune Kim, Taekin Kim, Jinhwan Na, Jaeyoung Lee, Seunghwan Jeong
2. Document editing system supporting online document creation with search function, *KR100143446*, 2022
Hojune Kim, Jinsu Choi

SCHOLARSHIPS

Chungsoo Overseas Scholarship Chungsoo Scholarship Foundation	2025 – 2027
Korea-Germany Junior Research Fellowship Support Max Planck POSTECH/KOREA	09/2024
Global Leadership Program Scholarship SNU	02/2024
Kwanjeong Undergraduate Scholarship Kwanjeong Educational Foundation	2021 – 2024